

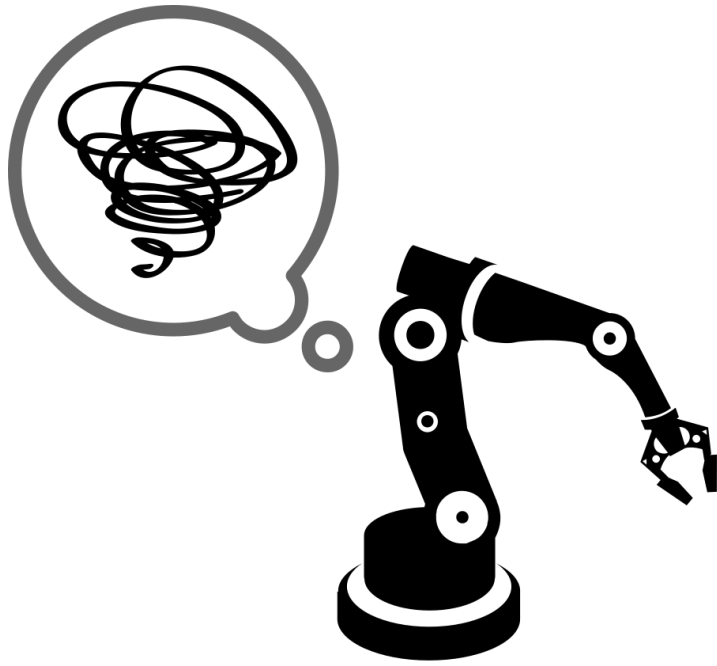
Communication



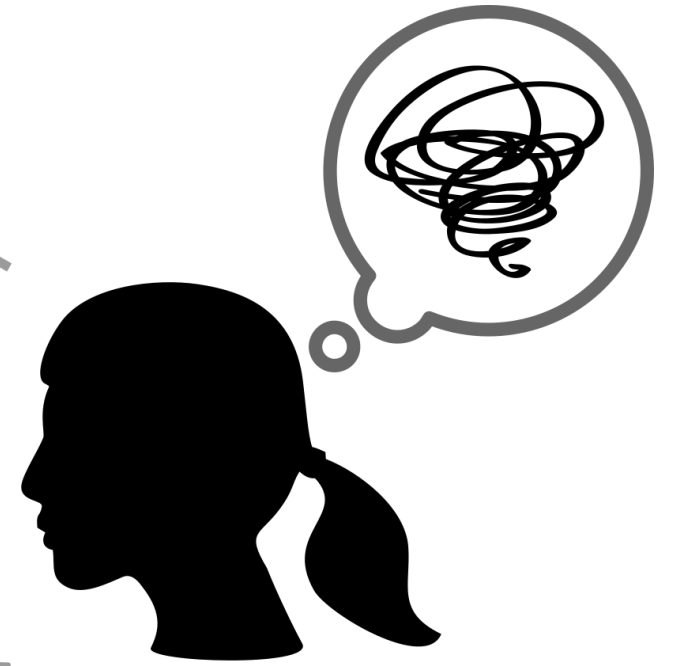
Dylan Losey

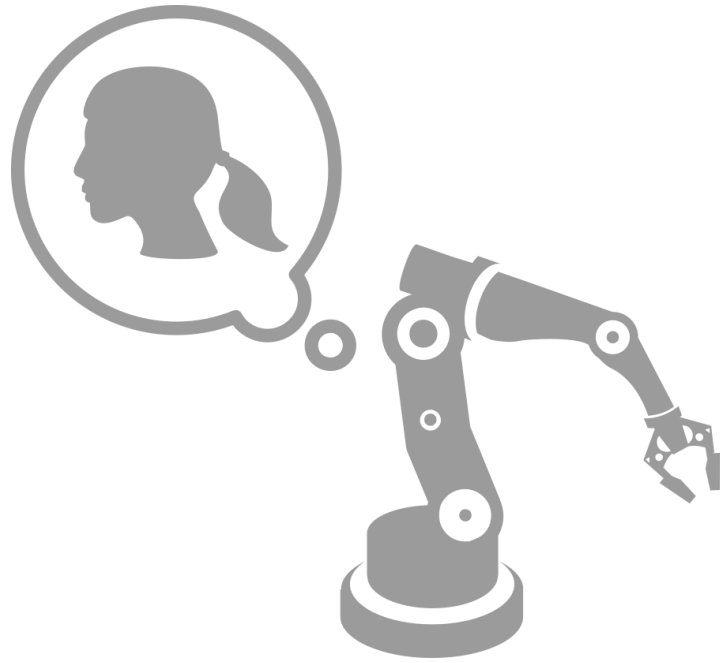
End-User Robot Programming using Mixed Reality

Samir Yitzhak Gadre, Eric Rosen, Gary Chien
Elizabeth Phillips, Stefanie Tellex, George Konidaris



*Human-Robot
Interaction*





*Human-Robot
Interaction*



This Lecture



- How can robots communicate with humans?
- How do we mathematically frame communication?

Communication

Robots can communicate to humans across multiple different modalities.

Language, AR/VR, forces, motions

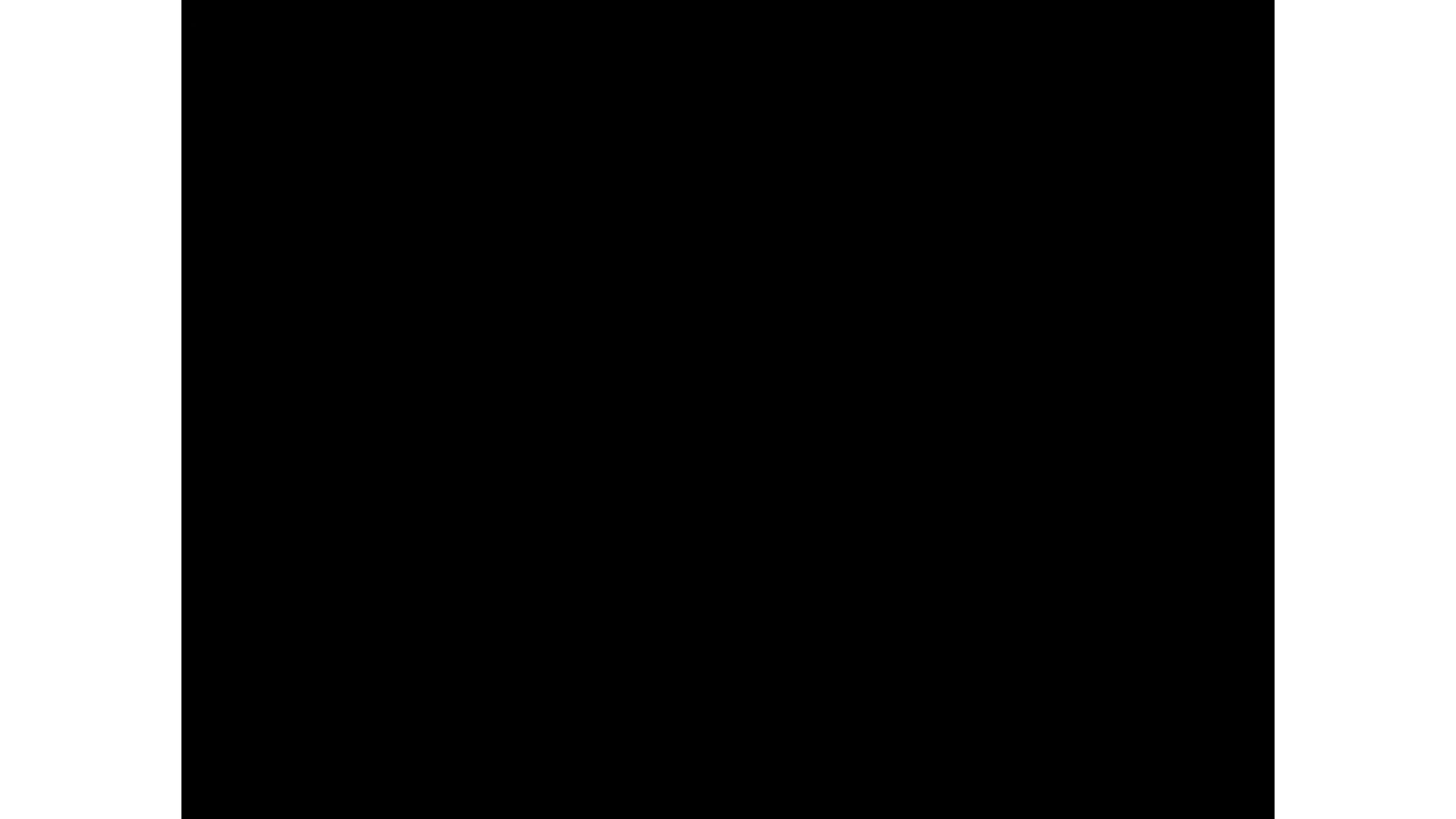
Through communication, robots seek to transfer knowledge about:

Goals, tasks, intent, regions of uncertainty, questions



What are some
popular
communication
methods?









Robot's Internal State

Consider a robot that is trying to **communicate** θ to the human

This θ is something the robot knows but the human does not

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Examples:

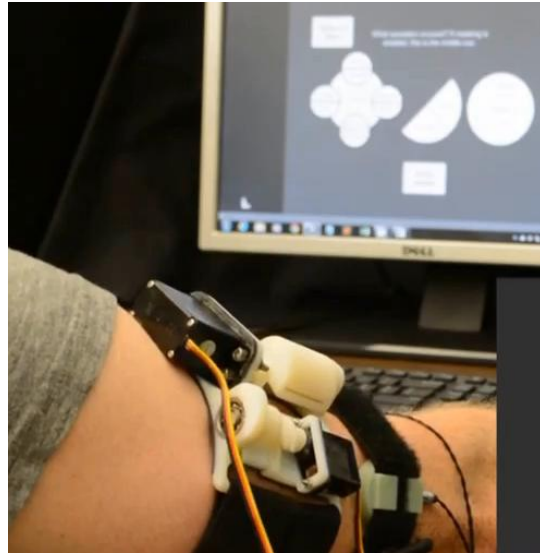
- The bounds of the robot's workspace (i.e., whether the robot can complete a task)
- The robot's intended motion (i.e., where the robot is going next)
- The robot's reward function (i.e., what the robot is optimizing for)
- What the robot has learned (i.e., whether or not the robot understands a task)

Robot's Internal State

Consider a robot that is trying to **communicate** θ to the human

This θ is something the robot knows but the human does not

Definition. More generally, let θ be the **robot's internal (e.g., hidden) state**.





How can robots
communicate with
their actions?



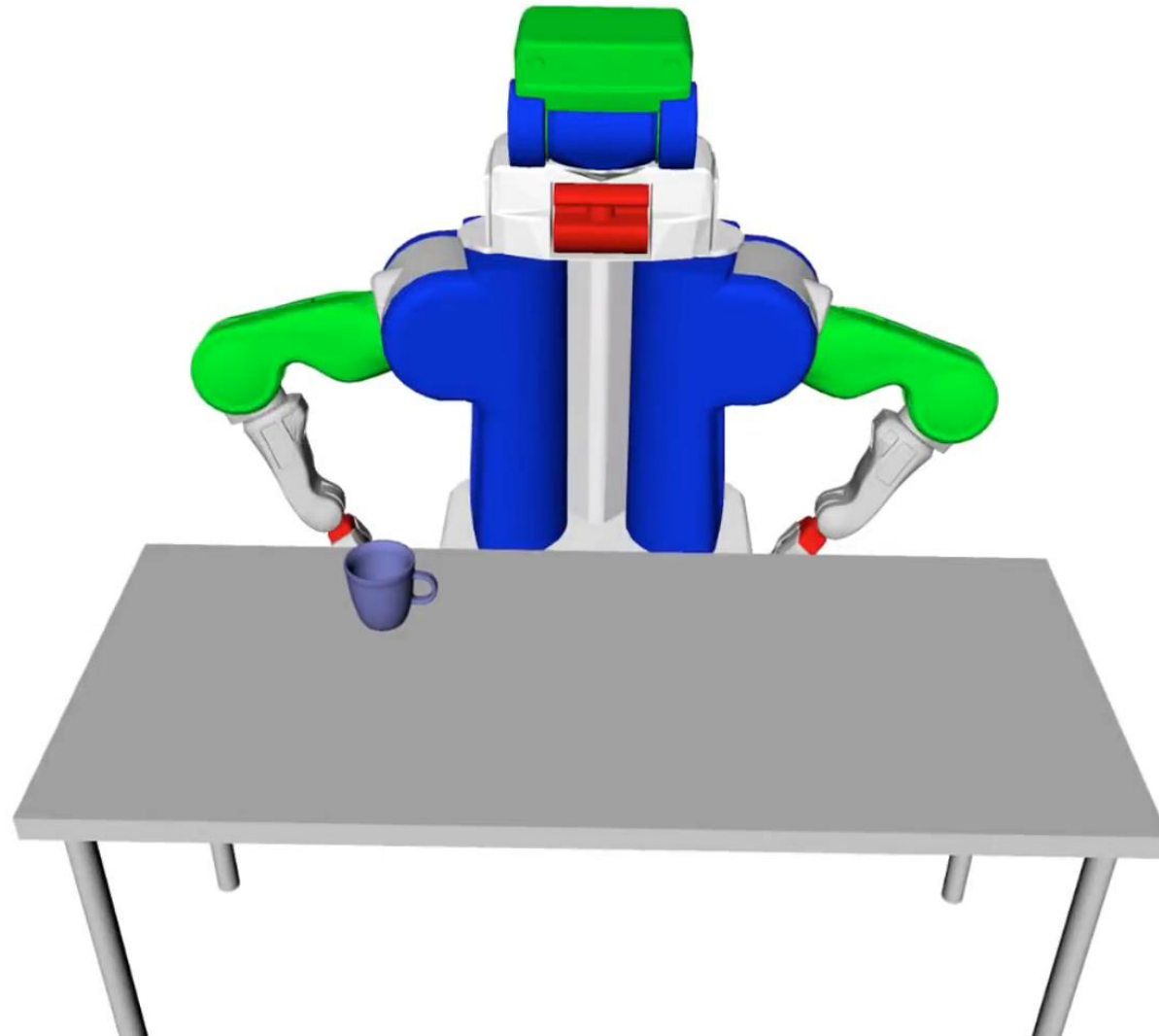
Legible Motion

Legible motion uses the robot's actions (e.g., the motion of the robot arm) to implicitly convey the robot's internal state.

Advantage: Does not require any additional hardware, seamless.

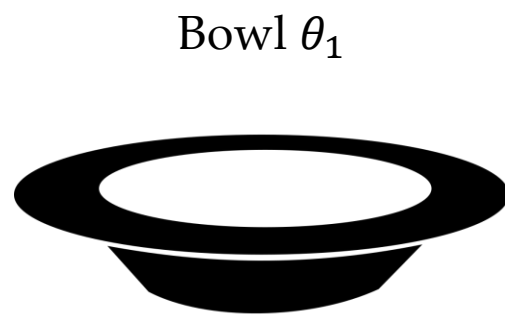
Disadvantage: Overloads actions, so that we need to balance between completing the task and communicating the robot's intent.

lift

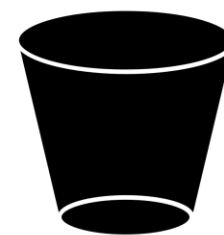




Robot with goal θ



Bowl θ_1



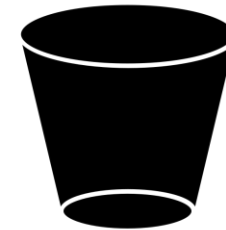
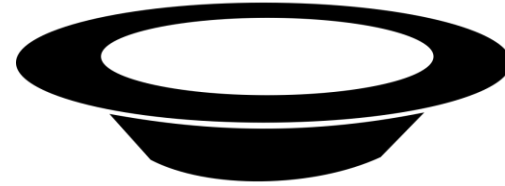
Cup θ_2

How could the robot move to θ in a way that conveys its internal state (goal)?



Robot with goal θ

Bowl θ_1



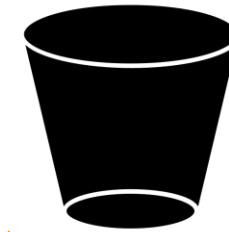
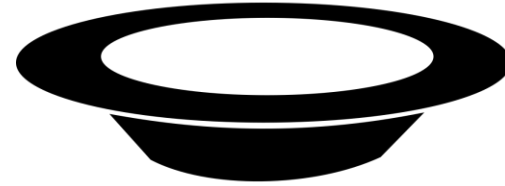
Cup θ_2

How could the robot move to θ in a way that conveys its internal state (goal)?



Robot with goal θ

Bowl θ_1



Cup θ_2



Legible Motion

When robots select **legible** actions (e.g., transparent, explainable, understandable), they **exaggerate** their motion to make θ clear to nearby humans

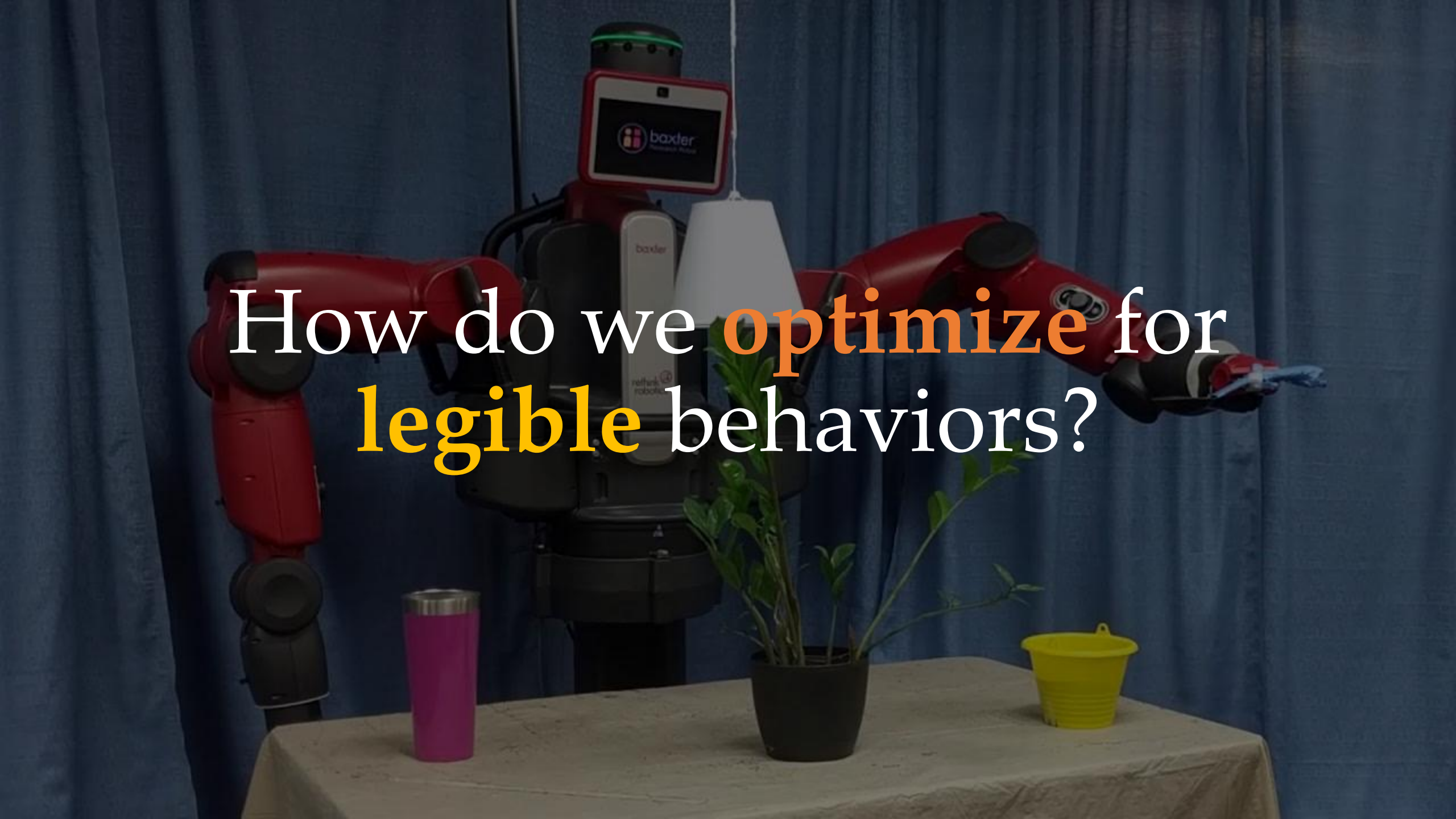
- Let θ be the robot's internal state (goal in our example)
- Let ξ be the robot's trajectory (sequence of states)

Legible Motion

When robots select **legible** actions (e.g., transparent, explainable, understandable), they **exaggerate** their motion to make θ clear to nearby humans

- Let θ be the robot's internal state (goal in our example)
- Let ξ be the robot's trajectory (sequence of states)
- Let $P(\theta \mid \xi)$ be a **model** of how the human interprets ξ to infer the robot's goal

Example. If we see the robot exaggerate to the right, we infer the robot wants the rightmost goal. In other words, $P(\theta_{right} \mid \xi) > P(\theta_{left} \mid \xi)$



How do we **optimize** for
legible behaviors?

Optimizing for Legibility

Legible robots select trajectories that best convey θ

$$\xi^* = \operatorname{argmax}_{\xi \in \Xi} P(\theta \mid \xi)$$

Robot's model of how human
interprets trajectory ξ

Optimizing for Legibility

Legible robots select trajectories that best convey θ

$$\xi^* = \operatorname{argmax}_{\xi \in \Xi} \frac{P(\xi \mid \theta)P(\theta)}{\sum_{\theta' \in \Theta} P(\xi \mid \theta')P(\theta')}$$

Alignment between ξ and θ

Alignment between ξ and the other possible goals

Optimizing for Legibility

Legible robots select trajectories that best convey θ

$$\xi^* \approx \operatorname{argmax}_{\xi \in \Xi} \frac{e^{\beta \cdot R(\xi, \theta)} P(\theta)}{\sum_{\theta' \in \Theta} e^{\beta \cdot R(\xi, \theta')} P(\theta')}$$

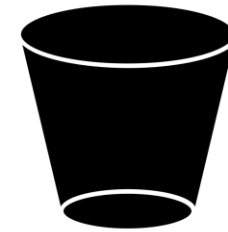
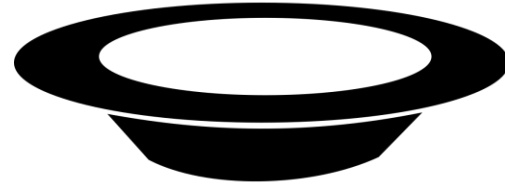
Human models the *robot* as an
approximately optimal agent

What happens if we **only** optimize for
communication, i.e., $P(\theta \mid \xi)$?



Robot with goal θ

Bowl θ_1



Cup θ_2

Optimizing for Legibility

Legible robots select trajectories that best convey θ ,
while also ensuring that they complete the intended task.

$$\xi^* = \operatorname{argmax}_{\xi \in \Xi} P(\theta \mid \xi)$$

$$\text{s. t. } \|\theta - \xi^T\| < \varepsilon$$

θ is the goal position, ξ^T is the final waypoint, and $\varepsilon > 0$ sets the allowable error

Optimizing for Legibility

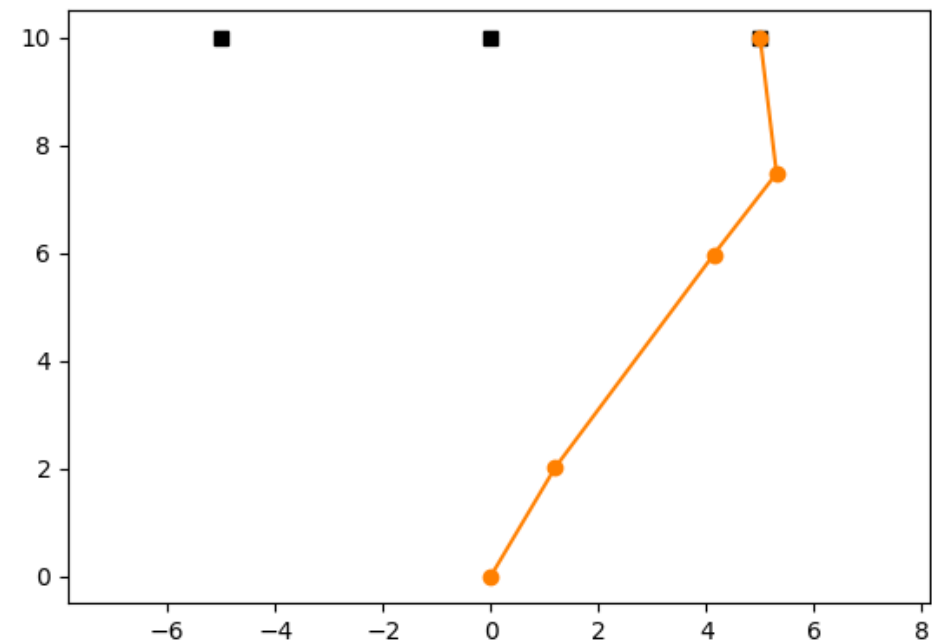
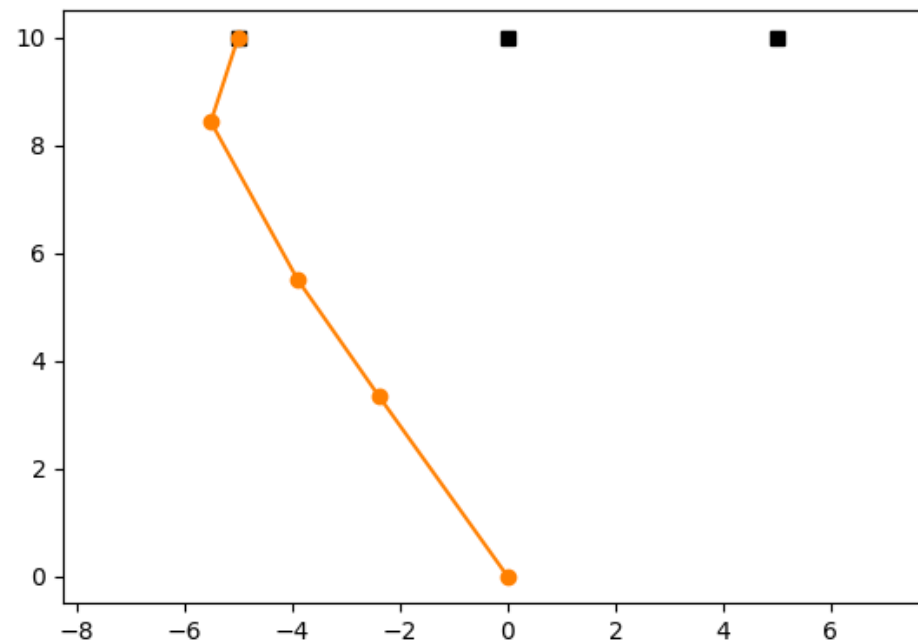
Legible robots select trajectories that best convey θ ,
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$$\xi^* = \operatorname{argmax}_{\xi \in \Xi} P(\theta \mid \xi)$$

← *Reveal θ*

$$\text{s. t. } \|\theta - \xi^T\| < \varepsilon$$

← *Reach θ*



Robot moving straight to its goal.



Robot using legible motion to convey its goal.

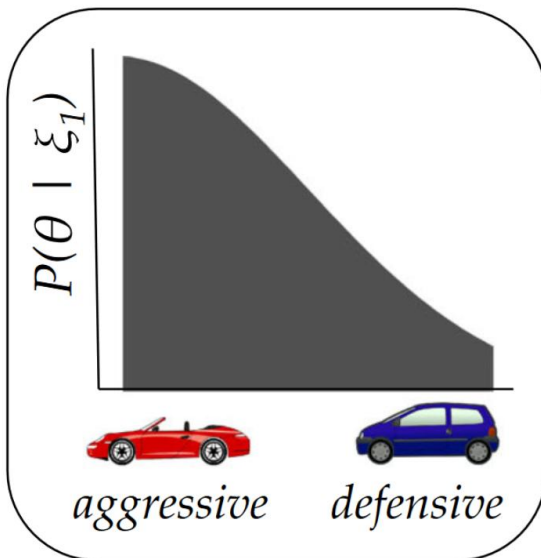
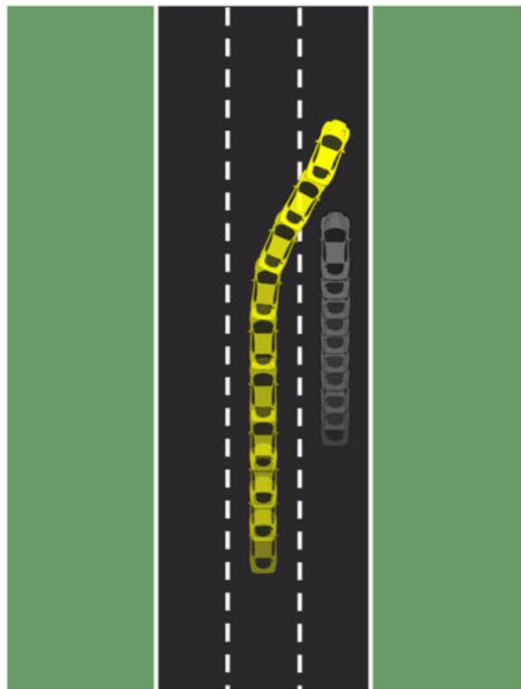




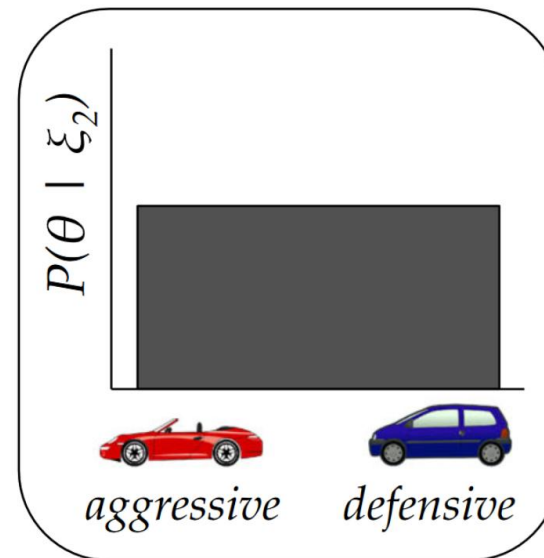
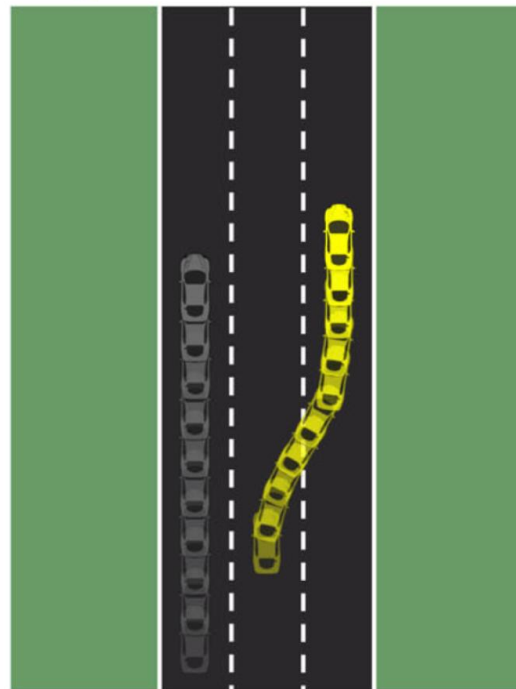
Can we extend
this to convey
more general
objectives?



ξ_1 (informative)



ξ_2 (uninformative)



Communicating Objectives

Let E be an **environment** we could show to the human.

- Each environment could be an image (GUI)
- Different environments show different human and robot behaviors
- We could simulate and sample these environments



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Let E be an **environment** we could show to the human.

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Let θ be the **parameters** of the reward function.

Let $R(E, \theta)$ be the **reward** for environment E under reward parameters θ .

Communicating Objectives

We will search through the set of possible environments to select the environment E that **best communicates** the robot's objective with parameters θ .

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$$E^* = \operatorname{argmax}_{E \in \mathbb{E}} P(\theta \mid E)$$

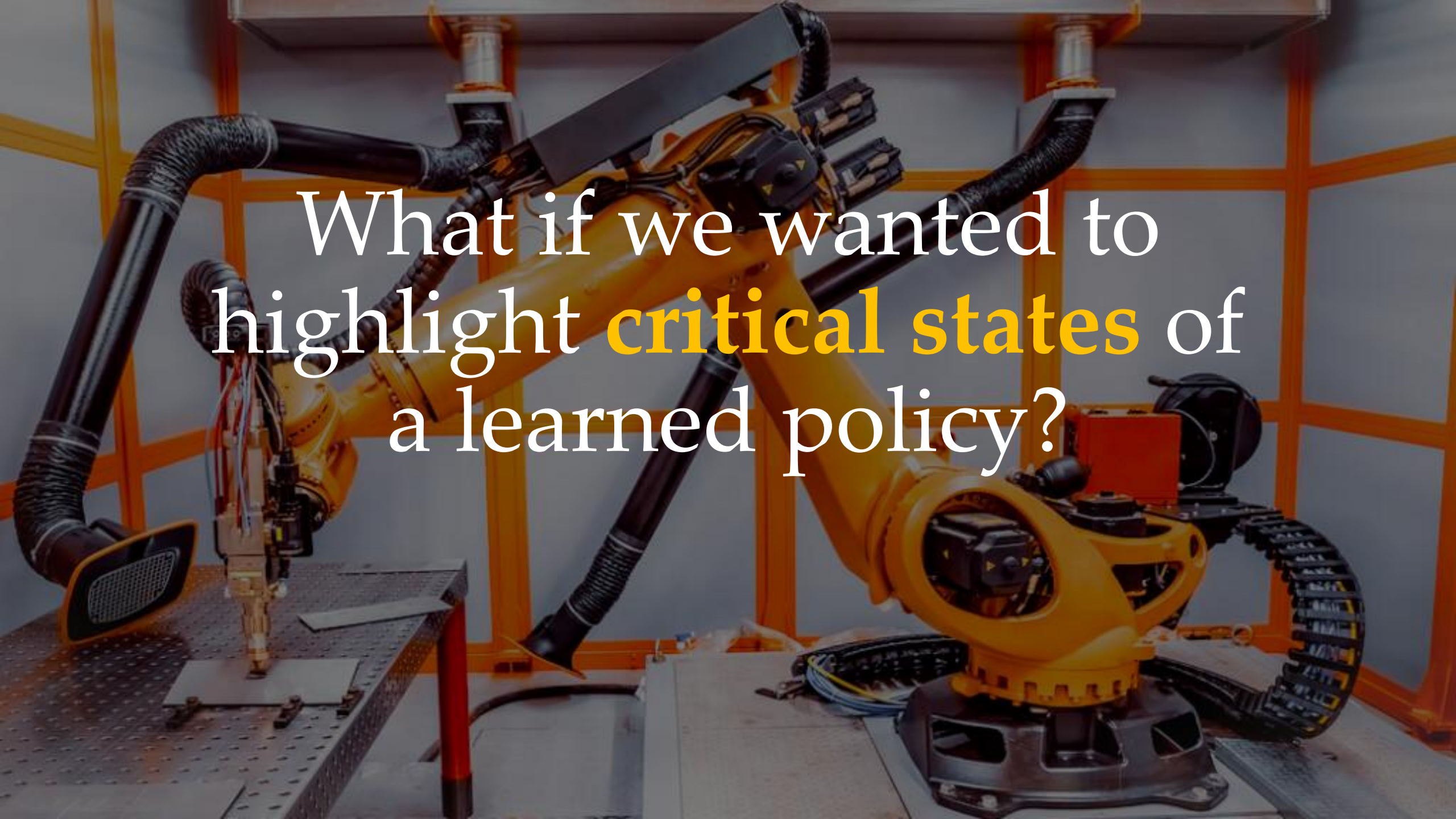
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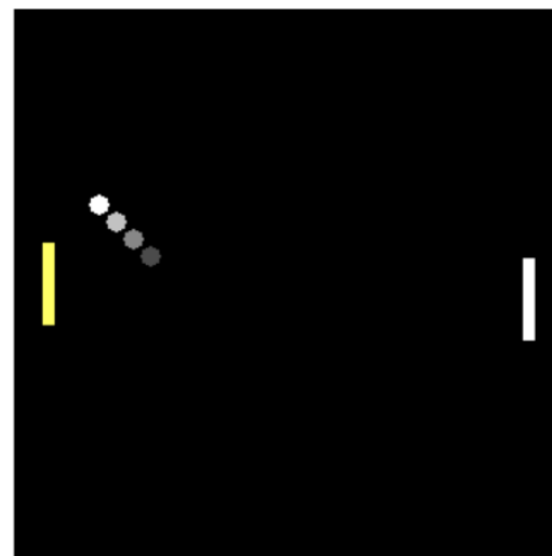
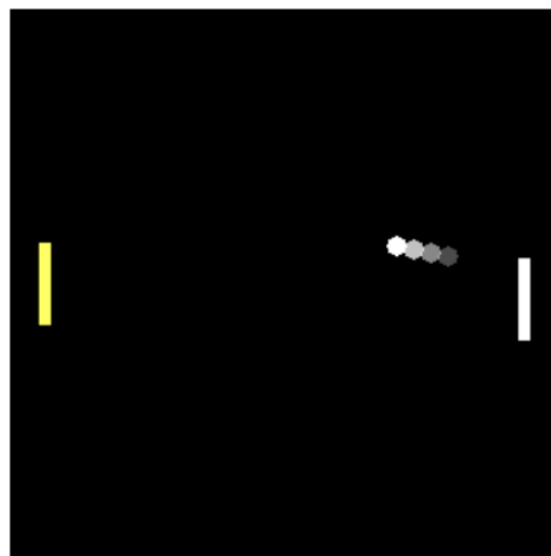
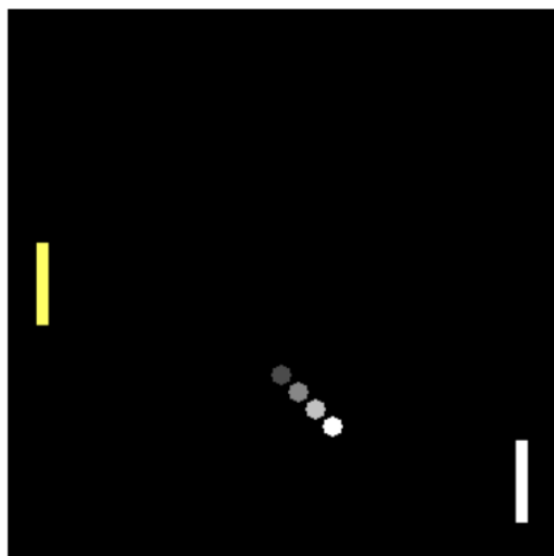
$$E^* = \operatorname{argmax}_{E \in \mathbb{E}} P(\theta \mid E)$$

$$P(\theta \mid E) \propto \frac{\exp R(E, \theta) \cdot P(\theta)}{\sum_{\theta' \in \Theta} \exp R(E, \theta') \cdot P(\theta')}$$

Same equations we used for legible motions, but now with E instead of ξ

The background image shows two yellow industrial robotic arms in a factory environment. One arm is on the left, positioned over a workbench with a perforated metal surface. The other arm is on the right, mounted on a base. Both arms have black flexible protective bellows on their joints. The background features an orange safety cage structure. The text is overlaid on the center of the image.

What if we wanted to
highlight **critical states** of
a learned policy?



Review

Q-function. $Q(s, a) \rightarrow \mathbb{R}$.

Cumulative discounted reward of taking action a at state s

$$Q(s^t, a^t) = r(s^t) + \gamma \cdot \max_a Q(f(s^t, a^t), a)$$

Reward at s

Long-term reward if we take action a

Can also write as $V(f(s^t, a^t))$

Review

Optimal policy. $\pi(s) \rightarrow a$.

Mapping from states to actions that maximizes total reward

$$\pi(s) = \max_a Q(s, a)$$

Take action that maximizes the Q-function

Critical States

- We want to show the robot's policy at a **few, important states**
- To identify these states we look for regions where: **acting randomly** will produce a much worse result than **acting optimally**

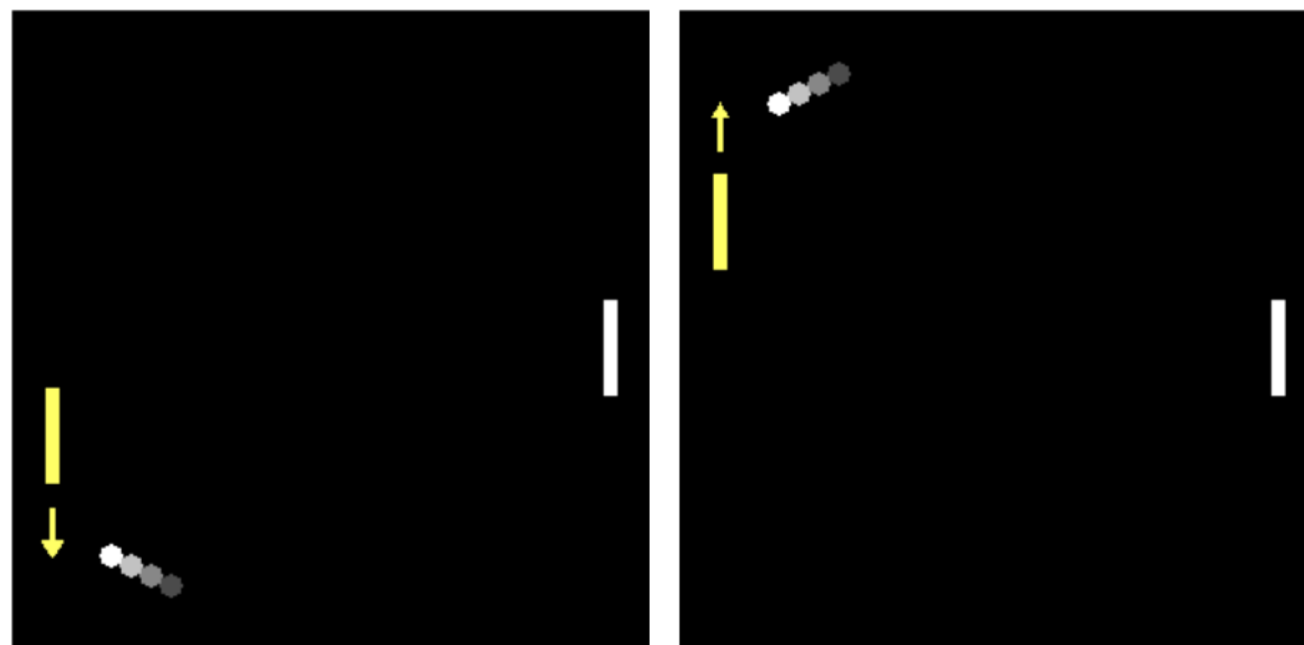
Critical States

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$$\max_a Q(s, a) - \frac{1}{N} \sum_{i=1}^N Q(s, a_i) > \varepsilon$$

Optimal action (policy)

Average Q value over all
 N possible actions



Takeaways

- **Communication** often makes HRI problems easier
- There are many ways to communicate, and the right one usually depends on **what we want to convey and the setting we are in**
- Communicating goals or objectives with **actions** is interesting because it does not require additional hardware / interruptions
- But doing this correctly relies on (1) an understanding of how the human interprets the robot's actions and (2) not overloading the robot's actions

This Lecture



- How can robots communicate with humans?
- How do we mathematically frame communication?

Next Lecture



- How can robots help humans be efficient teachers?